Arpan Pathak

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SUMMARY

Systems Software Engineer with ***8 years*** architecting high-performance, low-latency systems at ***Amazon***, ***Microsoft***, and ***Oracle***: database internals, kernel-level networking, distributed systems, and real-time ML inference. Deep expertise in ***Rust***, ***C++***, ***Go***, and ***C***: lock-free concurrency, memory safety, eBPF/Cilium, and GPU-accelerated computing. Shipped measurable wins: ***~ 40% lower P99 latency***, ***2x throughput***, ***45,000 ops/s*** throughput gains, ***5,000 QPS*** inference serving, and ***$5.4M/month*** revenue impact.

PERSONAL PROJECTS

Driving-CivicSense (Edge AI Vision) | github.com/arpanpathak/driving-civicsense-vision-model | 2026

- On-device AI perception system built in ***Rust*** for intersection discipline and lane awareness, running entirely on-device (***< 50ms inference latency***, no cloud dependency) for aftermarket dashcam and AR-glasses form factors.
- Fine-tunes ***YOLOv8/v11*** and custom ***CNN*** architectures; applies model pruning and quantization to fit tight memory and power budgets on low-power ARM single-board computers (Raspberry Pi Zero, Raspberry Pi 5, and other low-power ARM processor boards).
- Trains on cloud GPU infrastructure with synthetic data augmentation; planning to release a public driving-behavior dataset on ***Hugging Face***.

Companion Book: Seeing Machines: Deep Learning & Computer Vision from Python to Bare Metal | arpanpathak.github.io/seeing-machines-book

Physics & Mathematics of Game Development in Rust (Book) | arpanpathak.github.io/bevy-physics-book | 2026

CUDA Kernels: GPU & Parallel Programming from First Principles (Book) | arpanpathak.github.io/gpu-parallel-book | github.com/arpanpathak/gpu-parallel-book | 2026

EXPERIENCE

Software Developer II, Microsoft, Redmond, WA — Jun 2025 - Aug 2026

- ***C++/C# to Rust Migration (Memory Safety)***: Spearheaded migration of the Information Protection service from ***unsafe C++*** and ***C#*** to ***Rust***, moving memory safety to compile time: eliminated use-after-free, buffer-overflow, and null-safety bugs in the migrated path. Delivered ***~ 40% lower P99 latency***, ***2x throughput***, ***3x lower memory*** per request, and ***6-figure annual cost savings***.
- ***Zero-Trust Kernel Networking (eBPF/Cilium)***: Architected zero-trust sovereign cloud networking with ***eBPF + Cilium*** network policies enforcing ***data residency*** and ***SecNumCloud*** compliance; authored ***Kubernetes operators (Kubebuilder)*** to reconcile policy state across AKS clusters in France and Germany.
- ***AI-Powered Policy Engine***: Built an AI engine that auto-generates Cilium policies and scans for misconfigurations: manual review cut from ***4 hours to near-real-time***, zero policy drift across sovereign regions.
- ***CI/CD & Observability***: Containerized microservices with blue-green deployments, ***Prometheus/Azure Monitor***, and automated validation for reliable multi-region releases.

  TechStack: *Rust, C++, eBPF, Cilium, Kubernetes (AKS), Kubebuilder, Prometheus, Linux*

Senior Software Engineer, Oracle Cloud Infrastructure, Seattle, WA — Oct 2024 - Mar 2025

- ***High-Performance Database Internals***: Architected low-latency internals for a proprietary database engine, optimizing ***lock-free data structures*** to cut contention in high-concurrency read-write paths: ***45,000 ops/s*** transaction throughput gain.
- ***Infrastructure-as-Code SDK***: Engineered a ***Terraform provider*** (developer-facing SDK) for Oracle Autonomous Database, replacing legacy control-plane tooling with ***$1.2M/month*** projected savings.
- ***Security & Key Management***: Integrated OCI Security Vault across control-plane services for secret management and automated key rotation.

TechStack: ***Rust, C++, Go, Terraform, PostgreSQL internals, OCI***

Software Development Engineer II, Amazon, Seattle, WA & Hyderabad, India — Mar 2021 - Oct 2024

- ***Real-Time ML Inference Engine***: Architected ***XGBoost + Deep Learning ensemble*** ranking for ads & deals across FreeVee, Twitch, Prime Video, and Amazon Retail: millions of events at ***100ms*** ***inference latency***, driving ***5%*** ***CTR lift***, ***30%*** more deal impressions, and ***$5.4M/month*** ***revenue growth***.
- ***BERT Inference Service***: Delivered ***BERT-based*** NLP inference handling ***5,000 QPS*** at peak, cutting support-ticket triage SLA from ***7 days*** ***to*** ***2 hours***.
- ***Distributed Systems & SDK Design***: Designed uniqueness-constraint indexing and an ***SDK*** for a multi-tenant NoSQL store using ***two-phase commit***, achieving ***serializable isolation*** at ***3K writes/second***: a developer-facing library used by multiple publisher teams.
- ***Event-Driven Ads (CDC)***: Delivered programmatic guaranteed ad deals via real-time ***change data capture (CDC)*** with ***< 50ms latency*** for auction and programmatic buying platforms.
- ***Big Data Platform***: Built ***1-petabyte*** data lake & warehouse with ***Apache Spark***, ***AWS Glue***, Kotlin Spark API, and a ***DSL execution engine*** for OLTP and BI use cases.

TechStack: ***Kotlin, Java, Python, PyTorch, XGBoost, Spark, Kafka, DynamoDB, Redshift, AWS***

Software Development Engineer, Razorpay, Bengaluru, India — Aug 2020 - Feb 2021

- Built UPI Payment Gateway Aggregator in ***Go*** + Protobuf processing ***1M+ hourly transactions*** with ***99.99%*** ***uptime*** at ***150ms*** ***P99*** under ***50,000 TPM*** peak load.

TechStack: ***Go, Redis, PostgreSQL, Protobuf, Kafka***

Software Engineer, Mindfire Solutions, Bhubaneswar, India — Aug 2018 - Feb 2020

- Delivered full-stack workflow tools and an ***AR-based product search microservice*** serving ***10,000+ daily queries***; cut dashboard latency from ***5 min to 1 min***.

TechStack: ***Java, Spring Boot, Angular, MySQL, MongoDB***

SKILLS

Languages: Rust, C++, Go, C, Python, Java, Kotlin, TypeScript, Shell Script

Systems Programming & Performance: Lock-Free Data Structures, Concurrency, Memory Safety, Memory Profiling, Cache Optimization, Performance Engineering, Low-Latency Systems, OS Internals, Computer Architecture

Networking & Protocols: TCP/IP, UDP, TLS, mTLS, DNS, HTTP/2, HTTP/3, QUIC, WebSockets, Socket Programming, Network Performance Optimization

Network & Cloud Security: eBPF, XDP, Cilium, Kernel-Level Networking, Packet Processing, VPC, Subnets, Routing, NAT, VPN, Peering, Load Balancing, Security Groups / Firewalls

Distributed Systems & Streaming: Distributed Systems (consensus, replication, sharding), Apache Kafka, Apache Flink, Apache Spark, Real-Time Data Pipelines, Change Data Capture (CDC), Microservices

APIs & Serialization: gRPC, REST API Design, Protobuf

ML Inference & AI: PyTorch, TensorFlow, XGBoost, Transformer Models (BERT), LLMs, ONNX Runtime, On-Device/Edge Inference, Model Pruning & Quantization, Computer Vision (YOLOv8/v11), ANN, CNN, NLP, Deep Learning, Recommendation Systems, MLOps

GPU & Parallel Computing: CUDA, CUDA-Oxide (Rust-to-CUDA), PTX, SIMT, GPU Kernel Programming, Parallel Computing, GPU-Accelerated Computing

Cloud & Infra: AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS, Bicep), Google Cloud, Kubernetes (EKS, AKS), Docker, Terraform, Kubebuilder, Istio, Kubernetes Gateway API

Observability: Prometheus, Grafana, Azure Monitor, AWS CloudWatch

Data & Storage: DynamoDB, PostgreSQL, Redis, MongoDB, MySQL, Amazon Redshift, Amazon S3

Core CS: Data Structures & Algorithms, System Design, Operating Systems, Linux, Computer Architecture

EDUCATION

Bachelor of Technology in Computer Science & Engineering
RCC Institute of Information Technology, Kolkata, India | 2018